

1. Administrative & Study Identification

Official Study Title: A Phase III, Randomized, Double-blind, Multicenter, Global Study of Rilvegostomig or Pembrolizumab in Combination With Platinum-based Chemotherapy for the First-line Treatment of Patients With Metastatic Squamous Non-small Cell Lung Cancer Whose Tumors Express PD-L1 **Study Title:** A Global Phase III Study of Rilvegostomig or Pembrolizumab Plus Chemotherapy for First-Line Treatment of Metastatic Squamous Non-small Cell Lung Cancer (NSCLC) **Short Title/Acronym:** ARTEMIDE-Lung02 **Protocol Number:** D702BC00001 **NCT Number:** NCT06692738 **Sponsor/Company Name:** AstraZeneca **Investigational Product:** Rilvegostomig (AZD2936) or Pembrolizumab **Phase of Development:** Phase III **Medical Monitor:** AstraZeneca Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To assess the efficacy and safety of rilvegostomig in combination with platinum-based chemotherapy compared with pembrolizumab in combination with chemotherapy as first-line (1L) treatment in participants with squamous metastatic Non-Small Cell Lung Cancer (mNSCLC) whose tumors express PD-L1 (TC $\geq 1\%$), specifically focusing on Overall Survival (OS) and Progression-Free Survival (PFS). **Secondary Objectives:** To evaluate landmark Overall Survival (OS) rates, landmark Progression-Free Survival (PFS) rates, and time to second progression or death (PFS2).

2.2 Background & Rationale While anti-PD-1 therapies such as pembrolizumab in combination with chemotherapy represent the standard of care for first-line treatment of mNSCLC, many patients eventually experience disease progression. Rilvegostomig is a novel bispecific antibody targeting both PD-1 and TIGIT immune checkpoints. This study evaluates whether dual immune checkpoint blockade with rilvegostomig combined with chemotherapy can improve anti-tumor efficacy and survival outcomes compared to the standard pembrolizumab plus chemotherapy regimen.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Active Comparator (Pembrolizumab) **Blinding:** Double-blind **Randomization:** Randomized, two-arm, parallel group **Trial Status:** Recruiting

3.2 Planned Enrollment & Duration Target **\$N\$:** 880 patients **Number of Sites:** Global, Multicenter **Estimated Study Start:** 18-Nov-2024 **Estimated Primary Completion:** 05-Feb-2029

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Histologically or cytologically documented squamous NSCLC, Stage IV (based on the American Joint Committee on Cancer Edition 8) not amenable to curative treatment. 2. Absence of documented tumor genomic mutation results from tests conducted as part of standard local practice in any actionable driver oncogenes for which there are locally approved targeted 1L therapies. 3. Provision of acceptable tumor sample to confirm tumor PD-L1 expression (TC $\geq 1\%$). 4. At least one lesion not previously irradiated that qualifies as a RECIST 1.1 target lesion at baseline and can be accurately measured (≥ 10 mm in the longest diameter; lymph nodes must have short axis ≥ 15 mm) with CT or MRI. 5. Adequate organ and bone marrow function.

4.2 Exclusion Criteria 1. Presence of small cell and neuroendocrine histology components. 2. Brain metastases unless asymptomatic, stable, and not requiring steroids or anticonvulsants for at least 7 days prior to randomization (minimum 2 weeks elapsed between end of local therapy and randomization). 3. Any prior systemic therapy received for advanced or mNSCLC, or any prior treatment with an anti-PD-1, anti-PD-L1, anti-TIGIT agent, or other anticancer therapy targeting immune-regulatory receptors. 4. History of another primary malignancy (except for malignancy treated with curative intent with no known active disease ≥ 2 years before first dose). 5. Active or prior documented autoimmune or inflammatory disorders requiring chronic treatment with steroids or other immunosuppressive treatment. 6. Active primary immunodeficiency/active infectious disease(s) or active tuberculosis infection.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Intravenous infusions of Rilvegostomig or Pembrolizumab (active comparator) administered in combination with platinum-based doublet chemotherapy, including carboplatin (ATC code: L01XA02, trade name e.g., Paraplatin) + paclitaxel or nab-paclitaxel. **Route of Administration:** Intravenous (IV). **Duration of Treatment:** Cycles administered until documented disease progression, unacceptable toxicity, withdrawal of consent, or study termination.

5.2 Concomitant & Prohibited Therapy Permitted: Routine supportive care medications per institutional guidelines. **Prohibited:** Other concurrent investigational agents, unapproved systemic anti-cancer therapies, and chronic use of

immunosuppressive doses of systemic corticosteroids (unless indicated for specific pre-existing conditions).

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, Tumor Imaging, PD-L1 Assessment, Safety Labs
Baseline	Day 0	Randomization, First Dose of Study Treatment (IV)
Interim	Every 3 Weeks $\pm$ 3	Dosing, Safety Labs, PK/PD sampling, Tumor Imaging (every 6-9 weeks)
End of Study	Follow-up	End of Treatment Visit, Safety Follow-Up (30-90 days post-last dose), Survival Follow-Up

7. Outcome Measures (Endpoints)

7.1 Primary Endpoints

- 1. Overall Survival (OS):** Defined as the time from randomization until the date of death due to any cause.
- 2. Progression-Free Survival (PFS):** Defined as the time from randomization until radiological progression per Response Evaluation Criteria in Solid Tumors Version 1.1 (RECIST 1.1) or death due to any cause (in the absence of progression).

7.2 Secondary Outcomes

1. Landmark Overall Survival (OS) rates.
2. Landmark Progression-Free Survival (PFS) rates.
3. Time to second progression or death (PFS2).

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring and grading according to the National Cancer Institute Common Terminology Criteria for Adverse Events (NCI CTCAE). **Serious Adverse Events (SAEs):** Investigator reporting to the Sponsor within 24 hours of awareness. **Data Monitoring Committee (DMC):** An Independent Data Monitoring Committee (IDMC) will periodically assess the safety and efficacy data to ensure participant safety and trial integrity.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) population for all primary efficacy outcomes; Safety Analysis Set for adverse event evaluations. **Sample Size Justification:** $N = 880$ powered to demonstrate a statistically significant and clinically meaningful improvement in PFS and OS. **Handling of Missing Data:** Participants without documented progression or death will be right-censored at the date of their last evaluable tumor assessment according to standard FDA/EMA oncology guidelines.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: Risk-based monitoring utilizing frequent onsite and remote source data verification. **Audit Trail:** Robust electronic Data Capture (eDC) system with full audit trails, strict eCRF locking processes, and systematic data clarification mechanisms.

11. Study Identifiers & Governance

Primary ID: D702BC00001 **Registry Number:** NCT06692738 **Posting ID:** 66339227367613570 **Sponsor/Lead Organization:** AstraZeneca **Study Title:** ARTEMIDE-Lung02 **Trial Status:** Recruiting **Official Regulatory Title:** A Phase III, Randomized, Double-blind, Multicenter, Global Study of Rilvegostomig or Pembrolizumab in Combination With Platinum-based Chemotherapy for the First-line Treatment of Patients With Metastatic Squamous Non-small Cell Lung Cancer Whose Tumors Express PD-L1

12. Clinical Framework (Oncology Specific)

Disease Area: Non-small Cell Lung Cancer **Condition Name:** Metastatic Non Small Cell Lung Cancer **Preferred UMLS Name:** Non-small Cell Lung Carcinoma **Diagnosis Threshold:** Histologically or cytologically confirmed Stage IV squamous disease, PD-L1 TC $\geq 1\%$ **Medical Methodology:** 1st Line (1L) Chemo-Immunotherapy Dual Checkpoint Blockade (PD-1/TIGIT) **Optimization Target:** Delayed disease progression and extended overall survival

13. Study Procedures & Methodology

Management Pathway: Standard oncological clinical pathway for IV systemic therapy **Education Protocol (HCP):** Investigator protocol training, specialized AE management for bispecific immune checkpoint inhibitors **Education Protocol (Patient):** Informed consent review, infusion schedules, immune-related adverse event (irAE) recognition **Quality Audit Mechanism:** Centralized imaging review for RECIST 1.1 compliance, safety data reconciliation

14. Observation & Follow-Up Schedule

Total Duration: Follow-up until Study Completion (Estimated Oct 2029) **Onsite Visit Cadence:** Every treatment cycle (e.g., Q3W) for dosing and safety evaluations **Remote Monitoring Frequency:** Follow-up contacts for survival status post-progression **Checklist**

Review Interval: Regular tumor assessments every 6-9 weeks during the active treatment phase

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				